Type MS · Areas of regions made of several pieces

Answer 1 of 6 · Question 11 part c · 7 marks

2023 JAN · Paper 02 · find part c in this mark scheme extract

Question number		
11 a	When $x = 0 \quad y = 3$	B1
	When $y = 0 \quad 0 = 4 - e^{2x} \Rightarrow \ln 4 = 2x \Rightarrow x = \ln 2$	M1 A1 (3)
b	$\frac{dy}{dx} = -2e^{2x}$	M1
	When $x = \ln 2 \quad \frac{dy}{dx} = -8$	M1
	$y - 0 = \frac{1}{8}(x - \ln 2) \Rightarrow y = \frac{1}{8}x - \frac{1}{8}\ln 2$	M1 A1 (4)
c	$\int_0^{\ln 2}\left(4 - e^{2x}\right)dx = \left[4x - \frac{1}{2}e^{2x}\right]_0^{\ln 2} = \left(4\ln 2 - \frac{1}{2}e^{2\ln 2}\right) - \left(0 - \frac{1}{2}e^{0}\right)$	M1M1M1
	$\frac{1}{2}(\ln 2)\left(\frac{1}{8}\ln 2\right) = \frac{1}{16}(\ln 2)^2$	M1M1
	$\frac{1}{8}\int_0^{\ln 2}\lvert x - \ln 2\rvert\,dx = \frac{1}{8}\left\lvert\frac{x^2}{2} - x\ln 2\right\rvert_0^{\ln 2} = \left\lvert\frac{(\ln 2)^2}{16} - \frac{(\ln 2)^2}{8}\right\rvert = \frac{(\ln 2)^2}{16}$	[M1M1]
	$4\ln 2 - \frac{3}{2} + \frac{1}{16}(\ln 2)^2 = 1.3$	M1 A1 (7)
		Total 14 marks

Type MS · Areas of regions made of several pieces

Answer 2 of 6 · Question 11 part b · 11 marks

2025 JUN · Paper 02 (regional R) · extract 1 of 3 · find part b in this mark scheme extract

11 (a)	$(p=)3e^{\frac{2}{3}}$ oe	B1 [1]
(b)	$\left(\frac{dy}{dx}=\right)e^{\frac{x}{3}} = e^{\frac{2}{3}}$ oe	M1 A1
	$(m_{norm}=)-\frac{1}{"e^{\frac{2}{3}}"}=\left(-e^{-\frac{2}{3}}\right)$ oe	M1
	$y-"3e^{\frac{2}{3}}"="-\frac{1}{"e^{\frac{2}{3}}"}"(x-2)$ oe $\left(y=-e^{-\frac{2}{3}}x+2e^{-\frac{2}{3}}+3e^{\frac{2}{3}}\right)$	M1
	$0-3e^{\frac{2}{3}}=-e^{-\frac{2}{3}}(x-2)$ oe	dM1
	$x=2+3e^{\frac{4}{3}}$	A1
	$(\text{Area of triangle}=)\ \frac{1}{2}\left("3e^{\frac{2}{3}}"\right)\left("2+3e^{\frac{4}{3}}"-2\right)\left(=\frac{9}{2}e^2\right)$	ddM1
	$\int_0^2 3e^{\frac{x}{3}}dx=\left[\frac{3e^{\frac{x}{3}}}{\frac{1}{3}}\right]_0^2$	B1 M1
	$\left(\frac{3e^{\frac{2}{3}}}{\frac{1}{3}}\right)-\frac{3e^0}{\frac{1}{3}}\quad\left(=9e^{\frac{2}{3}}-9\right)$	dM1
	$=\frac{9}{2}e^2+9e^{\frac{2}{3}}-9$	A1cso
ALT for ddM1	$\int_2^{"2+3e^{\frac{4}{3}}"}(-e^{-\frac{2}{3}}x+2e^{-\frac{2}{3}}+3e^{\frac{2}{3}})dx=-e^{-\frac{2}{3}}\frac{x^2}{2}+2e^{-\frac{2}{3}}x+3e^{\frac{2}{3}}x$	
	$=\left[-e^{-\frac{2}{3}}\left(\frac{"2+3e^{\frac{4}{3}}"^2}{2}\right)+2e^{-\frac{2}{3}}\left("2+3e^{\frac{4}{3}}"\right)+3e^{\frac{2}{3}}\left("2+3e^{\frac{4}{3}}"\right)\right]$	ddM1
	$-\left[-e^{-\frac{2}{3}}\left(\frac{"2^2}{2}\right)+2e^{-\frac{2}{3}}(2)+3e^{\frac{2}{3}}(2)\right]$	
		[11]
		Total 12 marks

Type MS · Areas of regions made of several pieces

Answer 2 of 6 · Question 11 part b · 11 marks

2025 JUN · Paper 02 (regional R) · extract 2 of 3 · find part b in this mark scheme extract

Part	Mark	Additional Guidance
(a)	B1	For the correct value of p.
(b)	M1	Differentiates to get an expression of the form $ae^{\frac{x}{3}}$ $a \neq 3$
	A1	For $e^{\frac{2}{3}}$ This must be an exact value, but isw any following decimals.
	M1	Finds the negative reciprocal of their gradient. This is not a completely dependent mark but must come from any attempt at differentiation.
	M1	For a complete unsimplified attempt (if using $y - y_1 = m(x - x_1)$ to find the equation of l, using their value of p and their gradient, which can be any changed gradient after finding an expression and value for $\frac{dy}{dx}$ If $y = mx + c$ is used, a complete attempt must involve correctly rearranging (using their values for p and their gradient) to find c. **Different to question 2, if the candidate is not working with correct values and the rearrangement to find c with exact values is not shown, this mark cannot be awarded**
	dM1	Sets their equation = 0 and attempts to find a value for x. Allow errors in rearrangement. Dependent on previous method mark. This must be an exact value, but isw any following decimals.
	A1	Finds the correct value of x
	ddM1	For correct substitution into the formula for the area of a triangle, using (their x-intercept – 2) and their answer for p Does not need to be simplified. Dependent on previous 2 method marks. Or for stating the integral, with their p and their x intercept as limits and correct integration of at least 2 terms. **We don't need to see substitution of limits for this mark.** Their integral must be of the minimum form $\int_2^{"f+ge^{\frac{h}{3}x}"} \left(ae^{-\frac{c}{3}x} + be^{-\frac{d}{3}} \right) dx$ $a,b,c,d,f,g,h \neq 0$ See notes for next B mark, can be awarded if limits are correct, within a combined integral.
	B1	For stating $\int_0^2 3e^{\frac{x}{3}}$ Note: some candidates are listing this integral by adding or subtracting it from the equation of the line. ie $\int_a^{"b"} \left(-e^{-\frac{2}{3}x} + 2e^{-\frac{2}{3}} + 3e^{\frac{2}{3}} \pm 3e^{\frac{x}{3}} \right) dx$ - mark B1 if limits are correct and B0 if not and potentially next M1 dM1 for anything that is obviously their line $\pm 3e^{\frac{x}{3}}$
	M1	Integrates to get an expression of the form $ae^{\frac{x}{3}}$ $a \neq 3$
	dM1	Substitution of correct limits into any changed expression, must see substitution of $x = 2$ and 0 (unless the 0 limit gives an evaluation of 0) Allow a correct final answer or a correct evaluation of this integral $9e^{\frac{2}{3}} - 9$ to imply this mark, if no errors in working. Examiners are not expected to evaluate to check incorrect expressions integrated. Dependent on previous method mark.
	A1cso	For a full and correct worked solution with no errors.

Type MS · Areas of regions made of several pieces

Answer 2 of 6 · Question 11 part b · 11 marks

2025 JUN · Paper 02 (regional R) · extract 3 of 3 · find part b in this mark scheme extract

Pearson Education Limited. Registered company number 872828
with its registered office at 80 Strand, London, WC2R 0RL, United Kingdom

Type MS · Areas of regions made of several pieces

Answer 3 of 6 · Question 7 part c · 6 marks

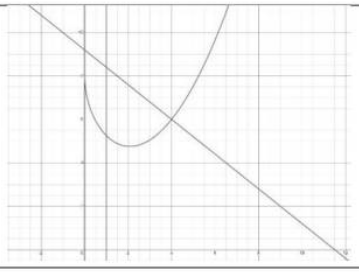
2024 JUN · Paper 01 · extract 1 of 5 · find part c in this mark scheme extract

7(a)	$(a=)\frac{4^2}{4}-3\sqrt{4}+8=6$	B1 cso [1]
(b)	$\left(\frac{dy}{dx}=\right)\frac{2x}{4}-3\times\frac{1}{2}\times x^{-\frac{1}{2}}$ oe $\left(x=4\Rightarrow\frac{dy}{dx}=\right)\frac{2\times4}{4}-3\times\frac{1}{2}\times4^{-\frac{1}{2}}=\frac{5}{4}\Rightarrow$ $M_n=-\frac{1}{"\frac{5}{4}"}$ $y-6="-\frac{4}{5}"(x-4)\Rightarrow 5y+4x-46=0$ *	M1 dM1A1 M1 (now matches ePen) M1A1 cso [6]
(c)	Area under C $\left(A_C=\int_1^4\left(\frac{x^2}{4}-3\sqrt{x}+8\right)\,dx\right)$ $=\left[\frac{x^3}{3\times4}-\frac{3\times x^{\frac{3}{2}}}{\frac{3}{2}}+8x\right]_1^4$ $=\left[\frac{4^3}{3\times4}-\frac{3\times4^{\frac{3}{2}}}{\frac{3}{2}}+8\times4\right]-\left[\frac{1^3}{3\times4}-\frac{3\times1^{\frac{3}{2}}}{\frac{3}{2}}+8\times1\right]$ $=\frac{61}{4}$ Area under the line $5\times0+4x-46=0\Rightarrow x(=11.5)\qquad A=\frac{1}{2}\times("11.5"-4)\times6=\frac{45}{2}$ **ALT** $\int_4^{11.5}\left(-\frac{4}{5}x+\frac{46}{5}\right)dx=\left[-\frac{4}{5\times2}x^2+\frac{46}{5}x\right]_4^{11.5}$ $=\left(-\frac{4}{5\times2}\times11.5^2+\frac{46}{5}\times11.5\right)-\left(-\frac{4}{5\times2}\times4^2+\frac{46}{5}\times4\right)$ Required area $=\left(\frac{61}{4}+\frac{45}{2}=\right)\frac{151}{4}(=37.75)$ oe	M1 M1 A1 M1A1 A1 [6]

Type MS · Areas of regions made of several pieces

Answer 3 of 6 · Question 7 part c · 6 marks

2024 JUN · Paper 01 · extract 2 of 5 · find part c in this mark scheme extract

ALT	Line - curve $\int_1^4\left(\left(-\frac{4}{5}x+\frac{46}{5}\right)-\left(\frac{x^2}{4}-3\sqrt{x}+8\right)\right)\mathrm{d}x$ or $\int_1^4\left(-\frac{x^2}{4}+3\sqrt{x}-\frac{4}{5}x+\frac{6}{5}\right)\mathrm{d}x$ $\left[-\frac{4}{5\times2}x^2+\frac{46}{5}x-\frac{x^3}{4\times3}+\frac{3\times x^{\frac{3}{2}}}{\frac{3}{2}}-8x\right]_1^4$ or $\left[-\frac{x^3}{4\times3}+\frac{3\times x^{\frac{3}{2}}}{\frac{3}{2}}-\frac{4}{5\times2}x^2+\frac{6}{5}x\right]_1^4$ $\left(-\frac{4}{5\times2}\times4^2+\frac{46}{5}\times4-\frac{4^3}{4\times3}+\frac{3\times4^{\frac{3}{2}}}{\frac{3}{2}}-8\times4\right)-$ $\left(-\frac{4}{5\times2}\times1^2+\frac{46}{5}\times1-\frac{1^3}{4\times3}+\frac{3\times1^{\frac{3}{2}}}{\frac{3}{2}}-8\times1\right)$ or $\left(-\frac{4^3}{4\times3}+\frac{3\times4^{\frac{3}{2}}}{\frac{3}{2}}-\frac{4}{5\times2}\times4^2+\frac{6}{5}\times4\right)-\left(-\frac{1^3}{4\times3}+\frac{3\times1^{\frac{3}{2}}}{\frac{3}{2}}-\frac{1}{5\times2}\times4^2+\frac{6}{5}\times1\right)$ $=\frac{127}{20}$ Area under the line $5\times0+4x-46=0\Rightarrow x\left(=11.5\right)\ 5\times y+4\times1-46=0\Rightarrow x\left(=\frac{42}{5}\right)$ $A=\frac{1}{2}\times\left("11.5"-1\right)\times"\frac{42}{5}"=\frac{45}{2}\quad A=\frac{1}{2}\times\left("11.5"-1\right)\times"\frac{42}{5}"=\frac{441}{10}$ **ALT** $\int_1^{"11.5"}\left(-\frac{4}{5}x+\frac{46}{5}\right)\mathrm{d}x=\left[-\frac{4}{5\times2}x^2+\frac{46}{5}x\right]_1^{"11.5"}$ $=\left(-\frac{4}{5\times2}\times"11.5"^2+\frac{46}{5}\times"11.5"\right)-\left(-\frac{4}{5\times2}\times1^2+\frac{46}{5}\times1\right)=\frac{441}{10}$ $\left(\frac{441}{10}-\frac{127}{20}=\right)\frac{151}{4}$	M1 M1 A1 M1 A1 A1
Useful diagram 		
		Total 13 marks

Type MS · Areas of regions made of several pieces

Answer 3 of 6 · Question 7 part c · 6 marks

2024 JUN · Paper 01 · extract 3 of 5 · find part c in this mark scheme extract

Part	Mark	Notes
(a)	B1	Correct substitution, no errors and shows $a = 6$
(b)	M1	For an attempt to differentiate the given function. Simplification not required. See General Guidance for the definition of an attempt. No power of x to increase.
	dM1	For substituting the value of $x = 4$ into their derivative. Can be implied by sight of $\frac{5}{4}$ Dependent upon the previous method mark.
	A1	For the correct gradient of $\frac{5}{4}$
	M1	For finding the negative reciprocal of their gradient.
	M1	For forming the equation of the normal, using the given value for a and their **changed** gradient (ie not the gradient of the tangent). This is not a dependent mark, but the candidate may not use the gradient of the tangent and the gradient used must come from some differentiation work. If the candidate uses $y = mx + c$, a fully correct rearrangement to find c must be shown (this can be implied by a correct c for their equation) and concluded with the equation of the line written.
	A1 cso	For the correct equation of the line, minimum steps shown, no errors or omissions.
(c)	M1	For an attempt to integrate the given function. See General Guidance for the definition of an attempt. In this question at least one term correctly integrated and no power of t to decrease. Terms do not need to be simplified. Limits don't need to be present or correct.
	M1	For substituting the correct coordinates into their **changed** expression, the correct way around and subtracting. At least one correct substitution in one term of each limit fully shown. Their limits must be correct.
	A1	For the correct area under the curve. A correct answer here will imply M1 M1 A1 if the integration step has been shown and first M1 awarded. Solutions where the integration step has not been shown will score M0 M0 A0.
	M1	For a correct method to find the area of the triangle using their value for the intersection of line $L$with the x-axis . or Fully correct integration for line and substitution of limits of their 11.5 (following any attempt to find where L crosses the x-axis) and 4. May be implied by correct area. Terms do not need to be simplified.
	A1	For the correct area of the triangle
	A1	For the correct area of $\frac{151}{4}$

Type MS · Areas of regions made of several pieces

Answer 3 of 6 · Question 7 part c · 6 marks

2024 JUN · Paper 01 · extract 4 of 5 · find part c in this mark scheme extract

ALT	M1	For an attempt to integrate either of the functions as given – there **must** be a subtraction sign if listed separately. Also award this mark for integrating $\int_1^4\left(-\frac{x^2}{4}+3\sqrt{x}-\frac{4}{5}x+E\right)\mathrm{d}x \quad E\neq 0$ (Combines the constant terms incorrectly) See General Guidance for the definition of an attempt. In this question at least one term correctly integrated and no power of t to decrease. Terms do not need to be simplified. Limits don't need to be present or correct.
	M1	For substituting the correct coordinates into their **changed** expression, the correct way around and subtracting. At least one correct substitution in one term of each limit fully shown. Their limits must be correct.
	A1	For $\frac{127}{20}$
	M1	For a correct method to find the area of the large triangle using their value for the intersection of line $L$with the x-axis . or Fully correct integration for line and substitution of limits of their 11.5 (following any attempt to find where L crosses the x-axis) and 4. May be implied by correct area. Terms do not need to be simplified.
	A1	For $\frac{441}{10}$
	A1	For the correct area of $\frac{151}{4}$

Where students have combined expression incorrectly as follows:

$\int_1^4\left(\left(\frac{x^2}{4}-3\sqrt{x}+8\right)\pm\left(-\frac{4}{5}x+\frac{46}{5}\right)\right)\mathrm{d}x$ If the constant term is unsimplified, the highest mark will be up to M1 M1 A1 M0 A0 A0 if the work is seen for the first 3 marks as described above.

If they have simplified the constant term to give eg

$\pm\left(\frac{x^2}{4}-3\sqrt{x}+\frac{86}{5}\right)\pm\frac{4}{5}x$ or $\pm\left(\frac{x^2}{4}-3\sqrt{x}+\frac{86}{5}\right)\pm\frac{86}{5}x$

The highest mark will be M1 M0 A0 M0 A0 A0 if the first two terms in the curve and any constant term are integrated as per conditions above.

Type MS · Areas of regions made of several pieces

Answer 3 of 6 · Question 7 part c · 6 marks

2024 JUN · Paper 01 · extract 5 of 5 · find part c in this mark scheme extract

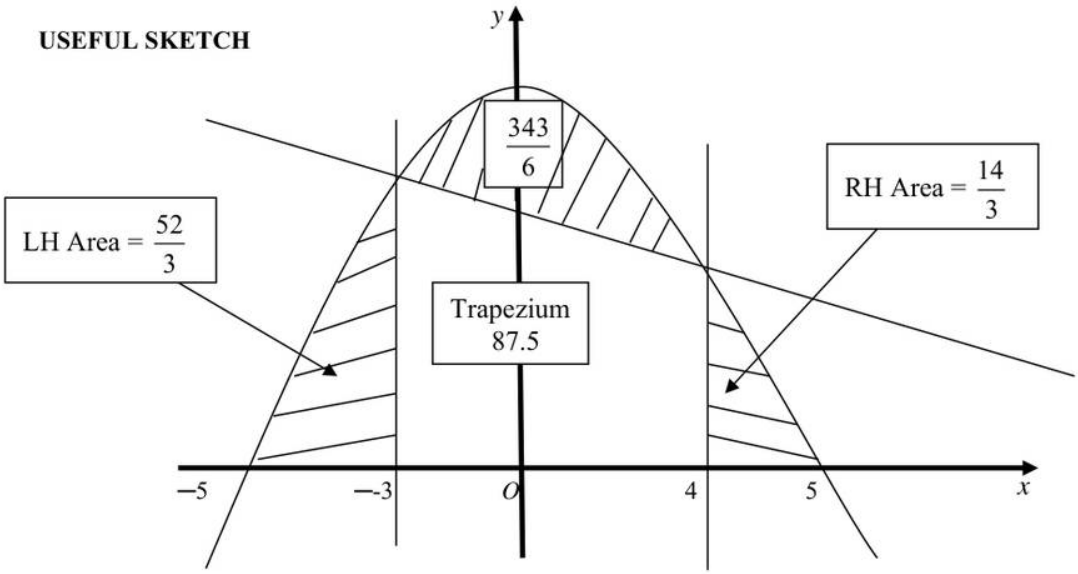
Question	Scheme	Marks

Type MS · Areas of regions made of several pieces

Answer 4 of 6 · Question 5 part b · 7 marks

2020 JAN · Paper 01 · extract 1 of 4 · find part b in this mark scheme extract

Number		
5 (a)	$25 - x^2 = 13 - x$ $x^2 - x - 12 = 0$ $(x - 4)(x + 3) = 0$ $A = (-3, 16)$ $B = (4, 9)$	M1 M1 A1 A1 (4)
(b)	$\int_{-5}^{5}(25 - x^2)\mathrm{d}x - \left[\int_{-3}^{4}(25 - x^2)\,\mathrm{d}x - \frac{1}{2}(16 + 9) \times 7\right]$ $\left[25x - \frac{x^3}{3}\right]_{-5}^{5} - \left\{\left[25x - \frac{x^3}{3}\right]_{-3}^{4} - 87.5\right\}$ $\left(\frac{250}{3} + \frac{250}{3}\right) - \left[\left(100 - \frac{64}{3}\right) - (-75 + 9) - 87.5\right]$ $\frac{219}{2} = (109.5)$	M1 A1 M1 A1 B1 M1 A1 (7)
	Alternative (b) $\int_{-5}^{5}(25 - x^2)\mathrm{d}x - \int_{-3}^{4}(12 - x^2 + x)\,\mathrm{d}x$ $\left[25x - \frac{x^3}{3}\right]_{-5}^{5} - \left[12x - \frac{x^3}{3} + \frac{x^2}{2}\right]_{-3}^{4}$ $\left(\frac{250}{3} + \frac{250}{3}\right) - \left(\frac{104}{3} + \frac{45}{2}\right)$ $\frac{219}{2} = (109.5)$	**[11]** M1 A1 M1 A1 A1 M1 A1 (7)



Type MS · Areas of regions made of several pieces

Answer 4 of 6 · Question 5 part b · 7 marks

2020 JAN · Paper 01 · extract 2 of 4 · find part b in this mark scheme extract

Part	Mark	Additional Guidance
(a)	M1	For setting the given equation of the curve = given equation of the line $25-x^2=13-x$ and attempting to form a 3TQ $x^2-x-k=0$ (k is an integer) Ignore the absence of = 0 if further work shows that they are attempting to solve a 3TQ = 0
	M1	For attempting to solve their 3TQ See general guidance for the definition of an attempt.
	A1	For **either** $(-3,\ 16)$ or $(4,\ 9)$
	A1	For **both** $(-3,\ 16)$ and $(4,\ 9)$
(b)		There are two ways to calculate this area. In each case; The first M mark is for a correct strategy (allow ft from (a) in their limits) The first A mark (M mark in Epen) is a fully correct strategy with correct limits The second M mark is for an attempt to integrate The second A mark is for a fully correct integration – ignore limits for this mark. The B mark (and A mark in Epen) is for the area of the trapezium of 87.5 seen anywhere. The third M mark is for substituting in their limits The final A mark is the correct answer only.
		Method 1 – Trapezium + two sides
	M1	For an attempt at the correct strategy to find the area. Allow for this mark a correct statement with using their limits correctly. This may well be seen at the end when they combine individual areas. $(A=)\frac{1}{2}('16'+'9')\times'7'+\int_{'4'}^{'5'}(25-x^2)\,dx+\int_{'-5'}^{'-3'}(25-x^2)\,dx$ OR $(A=)\int_{-3}^{4}(13-x)\,dx+\int_{'4'}^{'5'}(25-x^2)\,dx+\int_{'-5'}^{'-3'}(25-x^2)\,dx$
	A1	Fully correct expression with correct limits. $(A=)\frac{1}{2}(16+9)\times7+\int_{4}^{5}(25-x^2)\,dx+\int_{-5}^{-3}(25-x^2)\,dx$ OR $(A=)\int_{-3}^{4}(13-x)\,dx+\int_{'4'}^{'5'}(25-x^2)\,dx+\int_{'-5'}^{'-3'}(25-x^2)\,dx$
	M1	For an attempt to integrate their expression for area. (Follow General Guidance for the definition of an attempt) Ignore limits for this mark
	A1	For a fully correct integrated expression for the Area with a correct expression for the trapezium (**Ignore limits for this mark.**) $\left[13x-\frac{x^2}{2}\right],\ \left[25x-\frac{x^3}{3}\right],\ \left[25x-\frac{x^3}{3}\right]$ OR $\frac{1}{2}(16+9)\times7,\ \left[25x-\frac{x^3}{3}\right],\ \left[25x-\frac{x^3}{3}\right]$
	B1	For the correct area of the trapezium of 87.5. Award wherever seen. $\frac{343}{6}$ seen implies B1 If not seen explicitly, this can be implied from a correct final answer. **This is an A mark in Epen**

Type MS · Areas of regions made of several pieces

Answer 4 of 6 · Question 5 part b · 7 marks

2020 JAN · Paper 01 · extract 3 of 4 · find part b in this mark scheme extract

	M1	For an attempt to substitute **their** limits into **their integrated** expression.
	A1	For the correct final area of $A = \frac{219}{2}$ oe
	Method 2 – Using the area under the whole curve between −5 and 5; minus the area of the curve between −4 and 3; plus the area of the trapezium	
	M1	For an attempt at the correct strategy to find the area Allow for this mark, the correct strategy with **their** limits This may well be seen at the end when they combine individual areas. $(A=)\int_{-5}^{5}(25-x^2)\,dx-\int_{-3'}^{'4'}(25-x^2)\,dx+\frac{1}{2}('16'+'9')\times'7'$ OR $(A=)\int_{-5}^{5}(25-x^2)\,dx-\int_{-3'}^{'4'}(25-x^2)\,dx+\int_{-3'}^{'4'}(13-x)\,dx$ OR $(A=)\int_{-5}^{5}(25-x^2)\,dx-\int_{-3'}^{'4'}(12+x-x^2)\,dx$
	A1	For the correct expression with correct limits $(A=)\int_{-5}^{5}(25-x^2)\,dx-\int_{-3'}^{'4'}(25-x^2)\,dx+\frac{1}{2}('16'+'9')\times'7'$ OR $(A=)\int_{-5}^{5}(25-x^2)\,dx-\int_{-3'}^{'4'}(25-x^2)\,dx+\int_{-3'}^{'4'}(13-x)\,dx$ OR $(A=)\int_{-5}^{5}(25-x^2)\,dx-\int_{-3'}^{'4'}(12+x-x^2)\,dx$
	M1	For an attempt to integrate their expression for area. (Follow General Guidance for the definition of an attempt) Ignore limits for this mark
	A1	For a fully correct integrated expression for the Area with a correct expression for the trapezium. Accept this seen as individual parts Ignore limits for this mark. $\left[25x-\frac{x^3}{3}\right]$, $\left[25x-\frac{x^3}{3}\right]$, $\left[13x-\frac{x^2}{2}\right]$ OR $\left[25x-\frac{x^3}{3}\right]$, $\left[25x-\frac{x^3}{3}\right]$, $\frac{1}{2}(16+9)\times7$ OR $\left[25x-\frac{x^3}{3}\right]$, $\left[12x+\frac{x^2}{2}-\frac{x^3}{3}\right]$
	B1	For the correct area of the trapezium of 87.5 $\frac{343}{6}$ seen implies B1 Award wherever seen. If not seen explicitly, this can be implied from a correct final answer. **This is an A mark in Epen**
	M1	For an attempt to substitute **their** limits into **their integrated** expression or individual parts..
	A1	For the correct final area of $A = \frac{219}{2}$ oe

Type MS · Areas of regions made of several pieces

Answer 4 of 6 · Question 5 part b · 7 marks

2020 JAN · Paper 01 · extract 4 of 4 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Areas of regions made of several pieces

Answer 5 of 6 · Question 9 part c · 8 marks

2024 NOV · Paper 02 · extract 1 of 3 · find part c in this mark scheme extract

number		
9 (a)	$\cos 2\theta = \{\cos\theta\cos\theta - \sin\theta\sin\theta\} = \cos^2\theta - \sin^2\theta = \cos^2\theta - (1-\cos^2\theta)$ $= 2\cos^2\theta - 1$ *	M1 A1*cso [2]
(b)	$\int_{\frac{\pi}{3}}^{\frac{3\pi}{4}} \cos 2\theta \{\mathrm{d}\theta\}$ $\left[\frac{\sin 2\theta}{2}\right]_{\frac{\pi}{3}}^{\frac{3\pi}{4}} = \frac{\sin 2\left(\frac{3\pi}{4}\right)}{2} - \frac{\sin 2\left(\frac{\pi}{3}\right)}{2} = -\frac{2+\sqrt{3}}{4}$	M1 M1M1A1 [4]
(c)	$2\cos^2\theta - 1 = -\cos\theta$ $2\cos^2\theta + \cos\theta - 1 = 0 \Rightarrow (2\cos\theta - 1)(\cos\theta + 1) = 0 \Rightarrow \cos\theta = \ldots$ $\theta = \frac{\pi}{3}$ $\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} -\cos\theta \, \mathrm{d}\theta$ $= \left[-\sin\theta\right]_{\frac{\pi}{3}}^{\frac{\pi}{2}}$ $= -\sin\frac{\pi}{2} - \left(-\sin\frac{\pi}{3}\right) \left\{= \frac{-2+\sqrt{3}}{2}\right\}$ $\frac{2+\sqrt{3}}{4} - \frac{2-\sqrt{3}}{2}$ $\frac{-2+3\sqrt{3}}{4}$	M1 M1 A1 M1 M1 M1 ddM1 A1 [8]
ALT for last 5 marks	$\int_{-\frac{\pi}{3}}^{\frac{\pi}{2}} (-\cos\theta - \cos 2\theta)\{d\theta\} \{+\} \int_{\frac{\pi}{2}}^{\frac{3\pi}{4}} (-\cos 2\theta)\{d\theta\}$ $= \left[-\sin\theta - \frac{\sin 2\theta}{2}\right]_{-\frac{\pi}{3}}^{\frac{\pi}{2}} \{+\} \left[-\frac{\sin 2\theta}{2}\right]_{\frac{\pi}{2}}^{\frac{3\pi}{4}}$ $= \left(-\sin\frac{\pi}{2} - \frac{\sin 2\times\frac{\pi}{2}}{2}\right) - \left(-\sin\frac{\pi}{3} - \frac{\sin 2\times\frac{\pi}{3}}{2}\right) \{+\} \left(-\frac{\sin 2\times\frac{3\pi}{4}}{2}\right) - \left(-\frac{\sin 2\times\frac{\pi}{2}}{2}\right)$ $= \frac{-4+3\sqrt{3}}{4} + \frac{1}{2}$ $= \frac{-2+3\sqrt{3}}{4}$	M1 M1 M1 ddM1 A1
		Total 14 marks

Type MS · Areas of regions made of several pieces

Answer 5 of 6 · Question 9 part c · 8 marks

2024 NOV · Paper 02 · extract 2 of 3 · find part c in this mark scheme extract

Part	Mark	Additional Guidance
(a)	M1	For $\cos^2\theta-\sin^2\theta=\cos^2\theta-\left(1-\cos^2\theta\right)$ or $\cos^2\theta-\sin^2\theta=\cos^2\theta+\cos^2\theta-1$
	A1*cso	Obtains the correct result with no errors seen. Must have $\cos 2\theta$ on the lhs in their working, does not have to be in the final line provided there is a chain of reasoning.
(b)	M1	For replacement of $2\cos^2\theta-1$ with $\cos 2\theta$,ignore limits
	M1	For an attempt to integrate to an expression of the form $\pm\frac{\sin 2\theta}{2}$
	M1	Correct substitution of the **given** limits to a **changed** expression in terms of $\sin 2\theta$ and subtracts the correct way around. A correct value from a correctly integrated expression scores this mark, no need to see explicit substitution
	A1	Correct answer. isw
(c)	M1	Correctly equates the two curves.
	M1	Rearranges their 3TQ = 0 and attempts to solve (see general guidance). Can be implied by their correct $\cos\theta$ values, $\frac{1}{2}$ **and** –1 from a correct 3TQ, if their 3TQ is not correct, a clear method must be shown.
	A1	Correct angle found. Ignore extra angles.
	M1	**Main**: For stating the correct integral with "correct" limits of their $\frac{\pi}{3}$ and $\frac{\pi}{2}$ $\pm\int_{\frac{\pi}{3}}^{\frac{\pi}{2}}-\cos\theta\,\mathrm{d}\theta$ and $\pm\int_{\frac{\pi}{3}}^{\frac{3\pi}{4}}(2\cos^2\theta-1)\mathrm{d}\theta$ (the later integral can be implied by later work) (their $\frac{\pi}{3}$ must be positive and < $\frac{\pi}{2}$) **ALT**: Correct integrals $\pm\int_{\frac{\pi}{3}}^{\frac{\pi}{2}}(-\cos\theta-\cos 2\theta)\{\mathrm{d}\theta\}$ and $\pm\int_{\frac{\pi}{2}}^{\frac{3\pi}{4}}(-\cos 2\theta)\{\mathrm{d}\theta\}$ oe
	M1 **A1 on Epen**	For a correct integration of **all** their integrals in $\cos 2\theta$ and/or $\cos\theta$ only. Ignore limits
	M1	For correct **explicit** substitution of their limits into **either changed** expression, subtracts the correct way around – substitution of 0 must be seen unless substitution into their changed function gives 0 Can be implied by $\pm\left(\frac{-2+3\sqrt{3}}{4}\right)$ from correct working
	ddM1	Correct strategy finding the required area: **Main**: For subtracting the modulus of their 2 integrals. oe. Either way around. **ALT**: Add the values of two positive area "$\frac{-4+3\sqrt{3}}{4}$" and "$\frac{1}{2}$" oe Dependent on previous **three** method marks
	A1	Correct answer, simplest form and exact area of R

Type MS · Areas of regions made of several pieces

Answer 5 of 6 · Question 9 part c · 8 marks

2024 NOV · Paper 02 · extract 3 of 3 · find part c in this mark scheme extract

Question number	Scheme	Marks

Type MS · Areas of regions made of several pieces

Answer 6 of 6 · Question 12 part c · 4 marks

2020 NOV · Paper 01 (regional R) · extract 1 of 2 · find part c in this mark scheme extract

Question number		
12 (a)	$(2\cos x = 0)$ $\quad (x =)\dfrac{\pi}{2}$ or 90 °	B1 (1)
(b)	$(2\cos x = 2\sin x)$ $\quad \tan x = 1$	M1
	$x = \dfrac{\pi}{4}$ or 45°	A1 (2)
(c)	$\int_{(0)}^{(\frac{\pi}{4})} (2\sin x)dx + \int_{(\frac{\pi}{4})}^{(\frac{\pi}{2})} (2\cos x)dx$	M1
	$\left[-2\cos x\right]_0^{\frac{\pi}{4}} + \left[2\sin x\right]_{\frac{\pi}{4}}^{\frac{\pi}{2}}$	A1
	$\left(-\sqrt{2}+2\right)+\left(2-\sqrt{2}\right) = 4-2\sqrt{2}$	dM1
	$= 4-\sqrt{8}$ cao	A1cao cso (4) [7]

Part	Mark	Additional Guidance
(a)	B1	For $\dfrac{\pi}{2}$ or 90 degrees. Can also be shown as a coordinate – ignore any incorrect y coordinate.
(b)	M1	For $\tan x = 1$
	A1	For $\dfrac{\pi}{4}$ or 45 degrees. Can also be shown as a coordinate – ignore any incorrect y coordinate.
(c)	M1	For both integrals correctly shown, with an addition sign between them. Limits need not be shown. Must be 2 integrals shown. Can't be shown as one integral with incorrect limits.
	A1	For both functions correctly integrated. Limits need not be shown.
	dM1	For their limits clearly and correctly substituted in or for the numerical expression(s) shown in the MS. If mark awarded for substitution, both integrated expressions must have both limits correctly substituted. 0 must be the lower limit on the first integral. Allow ft of their $\dfrac{\pi}{4}$ (must be the upper limit on the first integral and the lower limit on the second and can be in degrees) and their $\dfrac{\pi}{2}$ (must be the upper limit on the second integral and can be in degrees).
	A1	cao cso A0 if degrees used in part c
		Note, can also be completed as either integral doubled – symmetry. M1 – correct integral stated with multiply by 2 evident or implicit later. A1 correctly integrated dM1 – as for main scheme, the multiply by 2 must be clearly shown. A1 as main scheme

Type MS · Areas of regions made of several pieces

Answer 6 of 6 · Question 12 part c · 4 marks

2020 NOV · Paper 01 (regional R) · extract 2 of 2 · find part c in this mark scheme extract